

MODULE 09

SQL Subqueries and Nested Queries

DBMS with SQL - Enterprise Learning Series

ARSHITH LMS | Professional Edition

SQL Subqueries and Nested Queries

What is a Subquery?

A SELECT nested inside another SQL statement.

Types: Scalar (one value), Row (one row),

Table (many rows), Correlated (references outer query).

WHERE Subquery

```
-- Students with above-average GPA
SELECT first_name, gpa FROM students
WHERE gpa > (SELECT AVG(gpa) FROM students);
```

IN vs EXISTS

```
-- IN
WHERE student_id IN (
  SELECT student_id FROM enrollments WHERE course_id = 101
);

-- EXISTS (faster for large data)
WHERE EXISTS (
  SELECT 1 FROM enrollments e
  WHERE e.student_id = s.student_id AND course_id = 101
);
```

Correlated Subquery

```
-- GPA above department average
SELECT s.first_name, s.gpa FROM students s
WHERE s.gpa > (
  SELECT AVG(s2.gpa) FROM students s2
  WHERE s2.department = s.department
);
-- Runs once per outer row
```

FROM Subquery

```
SELECT dept, avg_gpa FROM (
  SELECT department AS dept, AVG(gpa) AS avg_gpa
  FROM students
  GROUP BY department
) AS stats
WHERE avg_gpa > 3.0;
```